Muhamad Enrinal Zulhimar

Senior Backend Engineer
Jakarta, Indonesia (UTC+7) · Open to relocation and remote
enrinaal@gmail.com | linkedin.com/in/enrinal | github.com/enrinal

Summary

Senior backend engineer with 8+ years building distributed booking and payment systems in Go and Java. Owns the core flight booking services at Tiket.com, which handle 50,000 to 75,000 flight bookings a day and 5x surges at holiday peaks. Known for making the booking path reliable, with idempotent booking, failure recovery, and health monitoring. Also builds the team's AI agent tooling, including an MCP server and a Claude Code plugin.

Experience

Tiket.com Jakarta, Indonesia
Senior Software Engineer II, Backend Jul 2026 – Present

- Designed and built server-side cart management for flight checkout so users can recover an expired cart in one tap and get a reminder when they abandon one, improving the look-to-book ratio by about 15%.
- Made booking idempotent across all 7 booking API versions to close a race condition that could book the same order twice with the airline, cutting duplicates from 100–200 a day to zero.
- Fixed a memory leak outside the Java heap that crashed every flight-cart pod about 12 times a month. The cause was HTTP responses that were never released and compression that fragmented memory. Crashes dropped to zero.
- Built booking health monitoring across 4 flight services that alerts on drops in supplier booking success and order creation, so the team detects booking incidents within 5 minutes.
- Built the flight team's AI agent toolkit and knowledge base, so engineers investigate incidents and run team workflows through AI agents. Both are used by every engineer on the flight team and listed under Projects.
- Led system design reviews, wrote architecture decision records, ran on-call and incident post-mortems, and interviewed backend candidates, in a team of 5 engineers.

Senior Software Engineer, Backend Aug 2022 – Jul 2026

- Led the consolidation of fragmented MongoDB databases into one cluster and redesigned the data model to remove cross-database joins, cutting p99 latency by 40% and database cost by about 30%. Later reused by other verticals.
- Led the Flight + Hotel package booking project across two independent booking systems. Built a saga orchestrator in Go with compensating transactions to prevent charged-but-unbooked orders.
- Split tightly coupled legacy flight services into separate services and standardized shared internal libraries across squads, so teams ship features in parallel and new engineers ramp up about 35% faster.
- Created a rule-based pricing engine for add-ons that lets product teams launch pricing changes without a deployment, lifting upsell conversion by about 13%.
- Cut Redis memory from 48% to 15% of capacity with Snappy compression, and prevented cache stampedes.
- Shipped real-time anomaly detection on Kafka that auto-blocks booking bursts from one device and IP.

SiCepat Express Jakarta, Indonesia
Software Engineer, Backend Jul 2021 – Jul 2022

- Implemented idempotent event processing for shipment tracking on Kafka and Redis, so no event is lost or applied twice and customers and merchants always see the correct shipment status.
- Built an OCR pipeline to verify merchant ID cards automatically, cutting merchant onboarding time by about 50%.

Warung Pintar Jakarta, Indonesia
Software Engineer, Backend Jan 2020 – Jul 2021

- Built Kafka-based notification and wallet services processing over 100,000 messages a day.

Angsur Bandar Lampung, Indonesia
Software Engineer Jan 2018 – Oct 2019

- Built credit underwriting APIs, cutting credit validation time by about 30%.

Projects

Flight Team AI Agent Toolkit Tiket.com
Internal developer tool · Claude Code plugin, MCP server, Python 2026

- An internal tool that lets AI agents read production systems and handle routine team work, used by every flight engineer. Before it, engineers investigating an incident opened Kibana, the production MongoDB explorer, Grafana, Slack and Jenkins by hand.
- Built an MCP server in standard-library Python that gives AI agents typed tools for all of those consoles. It signs in with the engineer's own browser session, so the team does not need shared API tokens.
- Packaged it with 22 agent skills and 9 agent rules as a Claude Code plugin that also installs into Cursor and OpenCode. The skills write Jira stories, RFCs and deployment tickets in the team format and find why a pull request failed CI.

Flight Team Knowledge Base Tiket.com
Internal knowledge base · Markdown, git, context engineering for AI agents 2026

- A shared knowledge base that every flight engineer and their AI agents read. Before it, knowledge about the team's services was spread across Confluence pages, Slack threads and individual engineers.
- Built a page for each of 21 flight services and a map of how a booking moves from search to ticket issuance, down to file and line, and ported on-call playbooks for every flight domain from Confluence.
- Added git hooks that check every note before commit, so the knowledge base stays consistent as engineers and agents write to it.

Kestrel, a native macOS client for Apache Kafka github.com/enrinal/kestrel
Personal open-source project, MIT license · Swift 6, SwiftUI, librdkafka · built with Claude Code 2026

- A Mac app for exploring Kafka clusters, so engineers can browse topics, records and consumer groups in a real window instead of a browser tab or the output of Kafka's shell scripts.
- Built it in Swift 6 and SwiftUI on top of librdkafka, with a command-line tool that shares the same engine. It decodes Avro through Schema Registry, shows consumer group lag, and manages Kafka Connect connectors.
- Made clusters with 2,000 topics usable by filtering the sidebar and limiting how many rows it draws, and shipped three releases as a self-contained disk image that runs on any Mac without Homebrew.

Skills

Languages: Go, Java (Spring Boot, Spring WebFlux, Project Reactor), Python, Swift, SQL
Data and messaging: MongoDB, PostgreSQL, Redis, Elasticsearch, Kafka, NATS
Infrastructure: GCP, Kubernetes, Docker, microservices, gRPC, GraphQL, Grafana, distributed tracing, SLO alerting
AI engineering: Claude Code, Cursor, MCP servers, agent skills, agentic workflows, context engineering

Education

B.Eng. Informatics Engineering, Institut Teknologi Sumatera, 2020, *Cum Laude* (GPA 3.66/4.00)
Awards: ICPC Asia Jakarta Regional finalist, ranked 21st

Teaching

Ruangguru Jakarta, Indonesia
Go and System Design Instructor, part-time 2022 – 2024

- Taught Go and system design in a program run with Indonesia's Ministry of Education, mentoring 200 university students across 5 batches.